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A transformer-based unified multimodal framework for Alzheimer's disease assessment

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ABSTRACT

In Alzheimer's disease (AD) assessment, traditional deep learning approaches have often employed separate methodologies to handle the diverse modalities of input data. Recognizing the critical need for a cohesive and interconnected analytical framework, we propose the AD-Transformer, a novel transformer-based unified deep learning model. This innovative framework seamlessly integrates structural magnetic resonance imaging (sMRI), clinical, and genetic data from the extensive Alzheimer's Disease Neuroimaging Initiative (ADNI) database, encompassing 1651 subjects. By employing a Patch-CNN block, the AD-Transformer efficiently transforms image data into image tokens, while a linear projection layer adeptly converts non-image data into corresponding tokens. As the core, a transformer block learns comprehensive representations of the input data, capturing the intricate interplay between modalities. The AD-Transformer sets a new benchmark in AD diagnosis and Mild Cognitive Impairment (MCI) conversion prediction, achieving remarkable average area under curve (AUC) values of 0.993 and 0.845, respectively, surpassing those of traditional image-only models and non-unified multimodal models. Our experimental results confirmed the potential of the AD-Transformer as a potent tool in AD diagnosis and MCI conversion prediction. By providing a unified framework that jointly learns holistic representations of both image and non-image data, the AD-Transformer paves the way for more effective and precise clinical assessments, offering a clinically adaptable strategy for leveraging diverse data modalities in the battle against AD.

1. Introduction

Alzheimer's disease (AD) [1] is an important global health issue, being the main cause of dementia in those who are 65 years old and above. Mild cognitive impairment (MCI) represents a transitional stage between healthy aging and dementia [2]. MCI can be further categorized into progressive MCI (pMCI) and stable MCI (sMCI). Individuals with pMCI would convert to AD after several years, whereas those with sMCI exhibit relatively stable symptoms. Unfortunately, AD currently lacks a disease-modifying therapy, and clinical medication studies have experienced a high rate of failures [2–4]. Numerous researchers have been compelled by these discouraging figures to suggest that an accurate and early diagnosis of AD is necessary for subsequent clinical interventions and delaying the emergence of cognitive symptoms. Recently, machine learning and deep learning models have exhibited considerable potential in the detection of AD [5–7]. By analyzing diverse data sources such as brain imaging, genetic markers, and clinical records, these models can identify patterns and markers that aid in the early diagnosis and prediction of AD. Their ability to process complex data and detect subtle patterns has the potential to significantly impact the accuracy and timeliness of AD detection, offering hope for earlier interventions and more effective treatment strategies.

Since structural magnetic resonance imaging (sMRI) can delineate morphological changes in brain images, various machine learning and

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deep learning methods have been developed to identify AD patients through sMRI images. In the conventional machine learning models [8,9], the sMRI data undergoes preprocessing to extract distinct features, followed by training a classifier for AD diagnosis and MCI conversion prediction. Nevertheless, these conventional machine learning-based approaches heavily depend on the quality of features extracted from sMRI images, assuming that the selected features encompass the most discriminative information. Convolutional neural networks (CNN) can generally automatically learn the informative features due to their robust feature extraction and adaptive learning capabilities [10]. Significant progress has been achieved by CNN in the multi-classification of natural images and in AD diagnosis [11,12]. Based on the reviewed studies, image data management approaches may be categorized into four main types based on the kind of features extracted: voxel-based, sliced-based, region-based, and patch-based. The voxel-based approach is the most direct analysis technique, aiming to identify voxel-wise disease-associated microstructures for AD diagnosis and MCI conversion prediction [13]. However, it involves high feature dimensionality, requires a lot of computer power and overlooks local information of neuroimaging patterns. The slice-based technique extractes features based on 2D slices from a 3D image, and offers a solution to avoid handling a large number of parameters during training, leading to simplified networks [14]. Nevertheless, it sacrifices spatial dependencies in adjacent slices. Compared to slice/voxel-based approaches, the region-based methods extract region features using segmented regions from brain images and subsequently input into CNN models for AD diagnosis and MCI conversion prediction [15,16]. However, these methods just focus on the regions of interest (ROI), and thus may not encompass all potential pathological regions throughout the entire brain. In patch-based models [17–19], the 3D patches extracted from the 3D brain images are fed into 3D CNN models for AD diagnosis and MCI conversion prediction. Patch-based approaches are sensitive to minor alterations and can alleviate the curse of dimensionality.

Multimodal-based analysis can provide complementary information to enhance the detection of AD compared with the single-modal methods [6,7]. Current multimodal frameworks for AD detection are mostly based on the non-unified way to fuse information from different modalities [20,21]. Data from different modalities are directed to distinct models, generating modality-specific features or predictions. Subsequently, a fusion module is applied to consolidate these modality-specific features or predictions for making final diagnostic decisions. A notable drawback of fusion methods is the division of the multimodal diagnostic process into two relatively independent stages: modality-specific model training and diagnosis-oriented fusion. This design has an obvious limitation: it impedes the encoding of interconnections and associations among different modalities [22]. Recently, transformer-based architecture [23,24] have exerted a profound impact on the realm of natural language processing (NLP) and computer vision. Transformers operate under minimal assumptions regarding the form of the input data, possessing the capability to acquire feature representations of superior quality from multimodal input data. Importantly, the transformer network has the ability to capture the holistic connection across sub-patch regions [25], offering a chance to create a cohesive but adaptable model for performing representation learning on multimodal clinical information.

The main contributions of this paper can be summarized as follows: (1)A unified framework is constructed to integrate sMRI, clinical and genetic data for AD diagnosis and MCI conversion prediction. (2)The Patch-CNN is introduced to capture the discriminative information within each patch in 3D medical images. (3)A transformer based framework is designed to capture the interconnections among modalities for AD diagnosis and MCI conversion prediction by jointly learning holistic representations of medical images and non-image information.

2. Method

2.1. Study sample

Data used in the preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). The ADNI was launched in 2003 as a public-private partnership, led by Principal Investigator Michael W. Weiner, MD. The primary goal of ADNI has been to test whether serial MRI, positron emission tomography (PET), other biological markers, and clinical and neuropsychological assessment can be combined to measure the progression of mild cognitive impairment and early Alzheimer's disease [26]. All methods of the study were performed in accordance with the institutional ethics approval. We obtained the baseline subjects from the ADNI 1/GO/2 dataset (n = 1740) as study population. Exclusion criteria for study sample were as follows: (1) MCI participants at baseline who were lost to follow-up during the followed 36-month (n = 55); (2) participants with other dementia diagnose or without a definitive diagnosis of AD (n = 22); (3) participants with T1-weighted MRI scan missing or poor image quality at baseline (n = 12). As a result, 319 AD, 921 MCI and 411 cognitively unimpaired (CN) individuals were available for analysis. In our research, three distinct data modalities, comprising clinical information, genetic data, and MRI images, were employed. Clinical information included demographics data, neuropsychological assessments and functional assessments (Table 1).

2.2. Data preprocessing

All sMRI scans within the dataset were acquired in the NIFTI format. To promote data harmonization across the various cohorts, a set of preprocessing operations was implemented in a identical fashion in our study [7]. The entire preprocessing pipeline can be divided into four stages. (1) The scan axes were recalibrated to conform to the standard orientation of MNI-152 space. (2) By employing an automated technique, a three-dimensional volume-of-interest was identified within the original sMRI, selectively containing regions exclusively composed of

Table 1
Demographic information of participants in our study sample.

Variable	CN(n = 411)	sMCI(n = 659)	pMCI(n = 262)	AD(n = 319)	Pvalue
Age(year)	74.729 ± 5.733	72.845 ± 7.301	74.010 ± 6.825	75.176 ± 7.544	<0.001
Gender					
female	203(49.4 %)	287(43.6 %)	104(39.7 %)	143(44.8 %)	0.083
male	208(50.6 %)	372(56.4 %)	158(60.3 %)	176(55.2 %)	
Race					
other	41(10 %)	49(7.4 %)	9(3.4 %)	25(7.8 %)	0.019
white	370(90 %)	610(92.6 %)	253(96.6 %)	294(92.2 %)	
Years of education	16.277 ± 2.736	16.070 ± 2.857	15.798 ± 2.771	15.107 ± 2.989	<0.001
Ever married					
no	128(31.1 %)	169(25.6 %)	48(18.3 %)	50(15.7 %)	<0.001
yes	283(68.9 %)	490(74.4 %)	214(81.7 %)	269(84.3 %)	
RAVLT immediate	44.254 ± 9.819	38.351 ± 11.046	27.500 ± 6.685	22.940 ± 7.612	<0.001
RAVLT percent forgetting	37.595 ± 25.333	52.493 ± 30.031	78.763 ± 26.428	88.728 ± 21.587	<0.001
Functional Activities Questionnaire	0.156 ± 0.613	1.793 ± 2.912	5.789 ± 4.918	13.082 ± 6.905	<0.001
Geriatric Depression Scale	0.754 ± 1.122	1.583 ± 1.407	1.676 ± 1.403	1.646 ± 1.450	<0.001

brain tissue. (3) The region of interest underwent a skull-stripping procedure, involving the elimination of the skull region. (4) A preliminary linear registration was conducted to align the skull-stripped brain with a standardized MNI-152 template. This step involved approximating a linear transformation matrix that facilitated the mapping from the original sMRI space to the MNI-152 space. Following the image registration, the intensities of all voxels were normalized. The preprocessing pipeline was implemented using the FSL software [27].

The non-image data encompassed both genetic and clinical information. Genome Wide Association Study (GWAS) data from ADNI was generated in accordance with previous descriptions and sourced from the ADNI database. Standard protocols, including quality control measures, population stratification, and logistic regression-based association analysis, were conducted using the PLINK software [28]. All of these aforementioned steps were implemented in the following manner [29]. Finally, a list of SNPs ranked based on p-value $< 1 \times 10^{-5}$ is generated, and expanded using a one-hot encoding approach. The ApoE4 genotype was dichotomously considered, distinguishing between individuals with at least one e4 allele and those without any e4 alleles. Clinical information including demographics, neuropsychological test results, and functional assessments were also collected. The detailed clinical information was presented in Table 1. Non-image variables with more than 30 % missing data were excluded. Missing value of remaining non-image variables were imputed using the random forest algorithm [30]. Imputation was separately performed for the training and testing sets using the imputation algorithm learned from the training set. This approach ensures that the testing set remains unaffected during the imputation process.

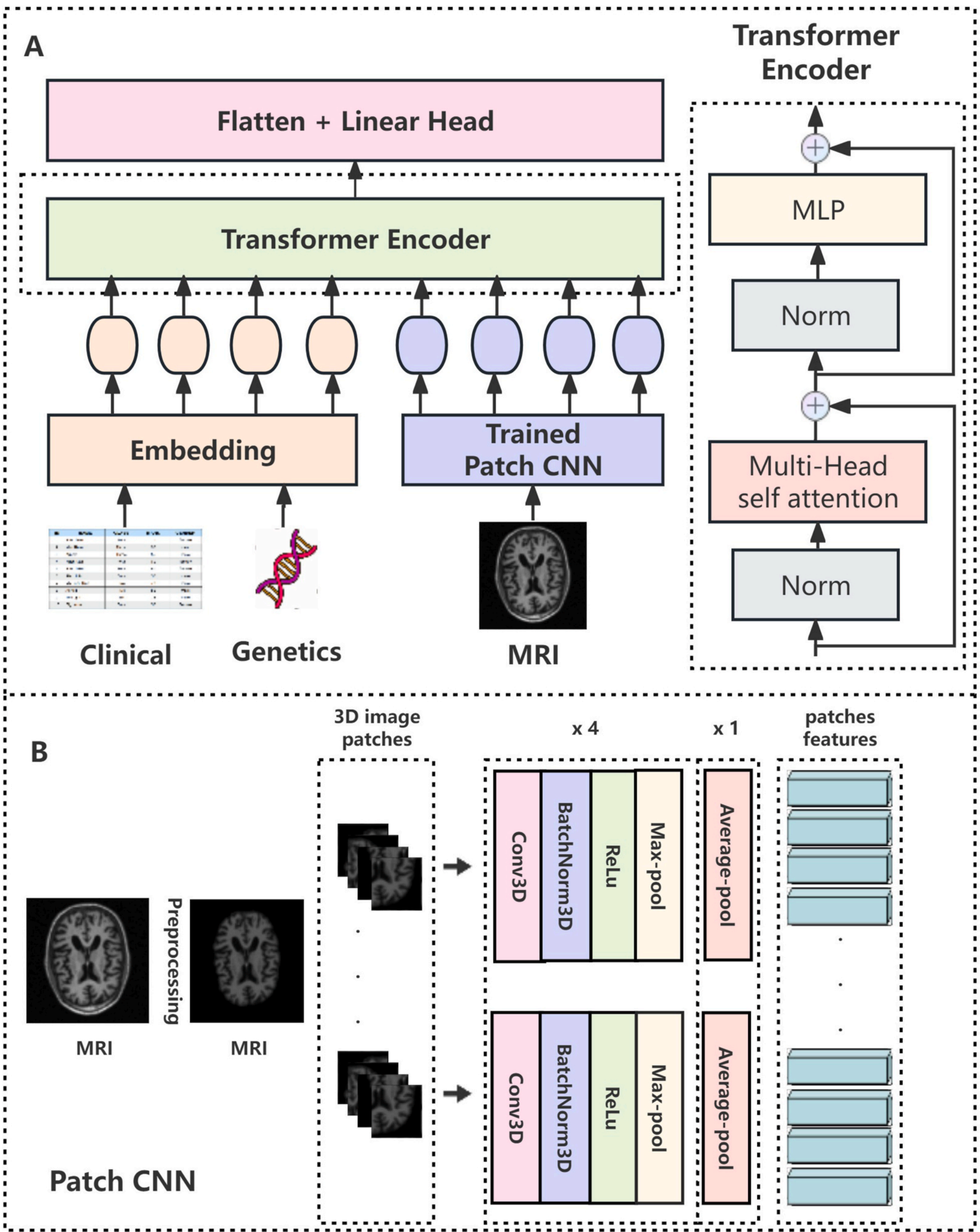

Fig. 1. Architecture of model construction
A, Overall workflow of AD-Transformer. The input data consist of two parts: the non-image (clinical and genetics) and image (MRI) data. Our model includes one embedding layer (liner projection for non-image data and trained CNN network for image data), one transformer layer and one linear classifier layer. B, Patch-CNN is a patch-level network used to learn the features of MRI images, each CNN branch composed of four convolutional blocks and a average pooling layer.

2.3. Outcome ascertainment

MCI was consistently defined across all phases of ADNI. To qualify as MCI, consistent criteria included the following [31]: First, the presence of reported cognitive complaints; Second, having a Mini-Mental State Exam (MMSE) score ≥24, a clinical dementia rating (CDR) equal to 0.5, and with preserved daily function; Third, individuals had to display measurable memory loss as determined by a Logical Memory test, adjusted for their years of education. Enrollment required individuals to exhibit impairment within the amnestic domain. The 921 MCI participants were subsequently categorized into pMCI and sMCI based on whether they would convert to Alzheimer's disease within 36 months following the initial assessment.

To meet criteria for AD, an individual was required to possess a CDR score of ≥0.5, an MMSE score of ≤26, an anomalous Logical Memory test, and meet the criteria for probable AD as per the NINCDS-ADRDA criteria at baseline [32].

2.4. AD-Transformer

We developed AD-Transformer, and forwarded multimodal data, including medical image and non-image information to the network for acquiring prediction results. During the training phase, we computed the binary cross-entropy loss by comparing the logits with the ground-truth labels. The primary advantage of this model is that the transformer encoder itself embeds both the image and non-image information and extracts interdependencies between input information.

AD-Transformer is a unified framework. Initially, two starting layers were constructed for embedding the tokens from the non-image and image data. Specially, a trained Patch-CNN subnetwork was initially constructed to extract discriminative features from the image data, and a linear projection block was employed to extract features from non-image data. Subsequently, a transformer encoder subnetwork was devised to learn mid-level representations among tokens. Ultimately, a classification head was utilized for AD diagnosis and MCI conversion prediction (Fig. 1).

The multimodal input data in the AD identification task consisted of two parts: non-image and image data. The Patch-CNN serves the purpose of acquiring more abstract features from the original image patches while simultaneously reducing the size of the feature map [33,34]. The 3D sMRI is transformed into a sequence of 3D patches with dimensions $x_p \in R^{H\times W\times D\times C}$, where H, W, and D are the height, width, and depth of the 3D patch respectively, and C denotes the number of channels. We fed the 3D patches into a Patch-CNN, resulting in a sequence of visual tokens. In our model framework, the deep Patch-CNN consists of m CNN branches, where each branch processes the patch at its corresponding location. Each CNN branch is composed of four convolutional blocks along with an average pooling layer. The convolutional block contained a 3D convolutional layer, a batch normalization layer, a ReLU activation function, and a 3D max-pooling layer. The dimensions of the convolutional kernel were 3 × 3 × 3. The filter size and stride size of max-pooling layer were 2 × 2 × 2 and (2,2,2), respectively. The number of convolution kernels from four convolutional blocks was set to 8, 16, 32, and 64 in order. After the average pooling, these patches (number = m) are subsequently projected to the embedding dimension h, forming a matrix $x_{image} \in R^{m\times h}$. Before we performed the formal training procedure, we pre-trained our Patch-CNN model, following the same approach as that employed for the baseline models. Simultaneously, non-image data (number = n) is also linearly projected to a dimension h, forming a matrix $x_{\text{non-image}} \in R^{n\times h}$. Specifically, min-max

scaling s was first employed to standardize each element of clinical data, and each standardized element was passed to a shared linear projection layer to obtain a sequence of embeddings. A similar procedure of tokenization was also performed on genetic data. The number for image and non-image embedding were 36 and 22, respectively. The image patch size was 48 × 48 × 48. The dimensions for image and non-image embedding were 64.

The projections of images and non-image are concatenated, forming a matrix $\in R^{(m+n)\times h}$. Positional encodings of the same dimension are added as learnable parameters to corresponding input tokens. The resulting embeddings matrix was then passed through a transformer encoder, which consists of normalization layers, multi-head self-attention mechanisms (MSA), and multi-layer perceptions. x is the input of the first attention block. The forward pass inside the attention layer was as follows:

$$Attention(Q, K, V) = softmax\left(\frac{QK^t}{\sqrt{d_k}}\right)V$$

$$head_i = Attention\left(QW_i^Q, KW_i^K, VW_i^V\right)$$

$$\mathrm{MSA} = concat(head_1, \cdots, head_h)W^O$$

$$x^l = x + MSA(\mathrm{Norm}(x))$$

where *Norm* denoted layer normalization, d_k was the dimension of the matrix.

W_i^Q, W_i^K, W_i^V were the linear transformation matrices of head i, i was the i th head in MSA. Subsequently, the output of attention layer was passed into the feed-forward network (FFN) [23]. The number of heads in MSA was set to 8, and the depth of encoder layers was 1.

$$x^{l+1} = x^l + FFN\left(Norm\left(x^l\right)\right)$$

Lastly, we flatten the matrix generated from the transformer block to obtain a holistic multimodal representation for disease prediction. This representation was passed to a two-layer MLP for AD diagnosis and MCI conversion prediction.

2.5. Baseline model

In our study, we compare our method with other deep neural networks that are also applied exclusively to image data, including 3D ResNet [35], 3D DenseNet [36] and VoxCNN [37]. In addition, a non-unified network (called AD-Net) was constructed for multimodal analysis. In AD-Net, a patch-CNN network was trained to extract patch features from whole image data, and the linear projection layer combined with classification layer was trained to extract features from non-image data. The derived image and non-image features were separately fed into two independent fully connected layers, resulting in two sets of vectors with the same dimension. The two sets of vectors were concatenated to create an embedding matrix. This embedding matrix was then fed into a two-layer MLP to generate the predicted label for the subject. All the MLP models were comprised of alternately stacked twice fully connected layers, ReLU activation functions, and dropout layers employed for the purpose of mitigating overfitting.

2.6. Model interpretation

We executed visualization using Grad-CAM [38] based on the individuals from the testing set. For image visualization, the target layer in Grad-CAM refers to the final convolutional layer in a CNN from which the gradients are computed. In general, we leverage backpropagation gradients to generate saliency maps for selected MRI images. The saliency maps disregard the influence of negative gradients and subsequently underwent min-max normalization. In addition, we also compute the backpropagation gradients of classification scores with respect to each image and non-image token. All the gradient values are normalized to [0, 1]. For multimodal analysis, the target layer in Grad-CAM refers to the FFN layer in a transformer. The gradient value of each token offers a cue for interpreting the models decisions, illuminating the specific tokens or positions within the input image and non-image data that influenced the models output. We also used SHapley Additive exPlanations (SHAP) technique to interpret the model. Initially developed for game theory applications, SHAP method can be applied in deep learning models by considering each voxel or network node as a unique feature [39]. Through assigning SHAP values to specific voxels or by mapping internal network nodes, SHAP offered insights into the contribution of individual features to the model's output.

3. Implementation details

Five-fold cross-validation was employed to assess the efficacy of a predictive model in the following manner: (1) The original dataset is randomly partitioned into five equal sized subsamples. From these subsamples, a single subsample was designated as the testing set, a single subsample was designated as the validation set, and the remaining three subsamples were utilized as training set. (2) Construction of the models was executed on each training and validation section, followed by an assessment of their performance on the test dataset. The training procedure was repeated five times, with each of the five subsamples used exactly once as the testing set. (3) The resulting performance metrics were subsequently presented as the mean value accompanied by the corresponding standard deviation. Throughout the training phase, we trained the model on the training set, assessed the model's performance on the validation set and computed the validation loss after each epoch. The model checkpoint with the lowest validation loss was preserved and subsequently evaluated on the testing set. We evaluated our model's performance on two distinct classification tasks: AD VS NC and pMCI VS sMCI. Subjects with AD/pMCI and NC/sMCI designations were regarded as positive and negative samples in the performance assessment, respectively. As these two tasks are highly correlated and the initial task is relatively easier, the trained model for AD diagnosis is transferred to initialize the training of model for MCI conversion prediction, facilitating faster convergence [31,40].

Our model is trained using the Adam optimizer for 50 epochs. The initial learning rate was set to 1×10^{-4}, while the weight decay was set to 1×10^{-2}. Besides, we used the warm-up and Cosine annealing algorithm to adjust the learning rate. Other hyperparameters of model were set as follows: batch size = 32, dropout rate = 0.3.

3.1. Performance metrics

We presented the classification performance by calculating the mean and the standard deviation over the five-fold cross-validation test (Tables 2 and 3). Four metrics, including accuracy, sensitivity, specificity, and the area under the curve (AUC) were computed. Besides, we also provided the ROC curve and AUC[95 % CI] based on all the test samples (Fig. 2). Delong's test was used to check the difference in AUCs between AD-Transformer and base models [41].

3.2. Statistic analysis

The baseline clinical variables including continuous variables and categorical variables were presented as mean ± standard deviation and number (percentage), respectively. Between group differences were analyzed with ANOVA for continuous variables and Chi-square test for categorical variables. Two-sided P-values less than 0.05 were considered statistically significant. All experiments run on CentOS with NVIDIA A800 GPU. Python 3.10 and PyTorch 1.11 was adopted to implement our model. Other statistical analyses were conducted in R software (4.0.2).

4. Result

4.1. Model performance

To assess the performance of our model for AD diagnosis and MCI conversion prediction, we compared our method with other deep learning methods. From Table 2, we can see that for the classification task of AD vs. NC, the average AUC of our AD-Transformer achieves 0.993, which is higher than that of image-only (ResNet: 0.870; DenseNet: 0.859; VoxCNN: 0.869) and non-unified (AD-Net: 0.974) methods, respectively. For the classification task of pMCI vs. sMCI, the average AUC of our method achieves 0.845, which is higher than that of image-only (ResNet: 0.742; DenseNet: 0.728; VoxCNN: 0.704) and non-unified (AD-Net: 0.828) methods, respectively. We also provided the ROC curve and AUC[95 % CI] based on all the test samples (Fig. 2). For AD diagnosis, the AD-Transformer exhibits the highest AUC[95 % CI] of 0.992 [0.988–0.996], compared to ResNet (0.868[0.841–0.894, $P < 0.001$), DenseNet (0.857[0.83–0.885], $P < 0.001$), VoxCNN (0.86 [0.834–0.886], $P < 0.001$), and AD-Net (0.973[0.962–0.984], $P < 0.001$). For MCI conversion, the AD-Transformer exhibits the highest AUC[95 % CI] of 0.837[0.809–0.865], compared to ResNet (0.742 [0.706–0.778, $P < 0.001$), DenseNet (0.722[0.685–0.758], $P < 0.001$), VoxCNN (0.702[0.664–0.74], $P < 0.001$), and AD-Net (0.815 [0.785–0.844], $P < 0.05$).

We also conducted an ablation study of the entire hybrid framework to assess its primary contributions. We compare the results of baseline Patch-CNN with those by gradually adding the non-image embedding block and transformer block. As shown in Table 3, the Patch-CNN network using image data achieves an average AUC of 0.901 and 0.741 for AD diagnosis and MCI conversion prediction. Further, we find

Table 2
Comparison between our proposed method and other deep learning methods.

Model	Accuracy	Sensitivity	Specificity	AUC
AD vs. NC				
3D ResNet	0.777 ± 0.027	0.723 ± 0.049	0.816 ± 0.066	0.870 ± 0.016
3D DenseNet	0.766 ± 0.033	0.777 ± 0.054	0.754 ± 0.083	0.859 ± 0.017
VoxCNN	0.780 ± 0.034	0.722 ± 0.051	0.824 ± 0.076	0.869 ± 0.016
AD-Net	0.915 ± 0.013	0.865 ± 0.008	0.953 ± 0.023	0.974 ± 0.011
AD-Transformer	0.959 ± 0.017	0.956 ± 0.011	0.961 ± 0.038	0.993 ± 0.005
pMCI vs. sMCI				
3D ResNet	0.701 ± 0.025	0.660 ± 0.047	0.719 ± 0.019	0.742 ± 0.018
3D DenseNet	0.671 ± 0.043	0.674 ± 0.045	0.669 ± 0.072	0.728 ± 0.016
VoxCNN	0.669 ± 0.045	0.549 ± 0.011	0.715 ± 0.059	0.704 ± 0.021
AD-Net	0.748 ± 0.061	0.694 ± 0.124	0.764 ± 0.131	0.828 ± 0.027
AD-Transformer	0.753 ± 0.026	0.745 ± 0.099	0.755 ± 0.071	0.845 ± 0.026

Table 3
The results of ablation study on the whole hybrid network.

Model	Accuracy	Sensitivity	Specificity	AUC
AD vs. NC				
Patch-CNN	0.845 ± 0.033	0.804 ± 0.052	0.876 ± 0.053	0.901 ± 0.015
AD-Transformer (no transformer block)	0.906 ± 0.029	0.888 ± 0.016	0.918 ± 0.055	0.959 ± 0.016
AD-Transformer	0.959 ± 0.017	0.956 ± 0.011	0.961 ± 0.038	0.993 ± 0.005
pMCI vs. sMCI				
Patch-CNN	0.716 ± 0.040	0.563 ± 0.029	0.775 ± 0.060	0.741 ± 0.025
AD-Transformer (no transformer block)	0.725 ± 0.045	0.695 ± 0.034	0.733 ± 0.072	0.804 ± 0.023
AD-Transformer	0.753 ± 0.026	0.745 ± 0.099	0.755 ± 0.071	0.845 ± 0.026

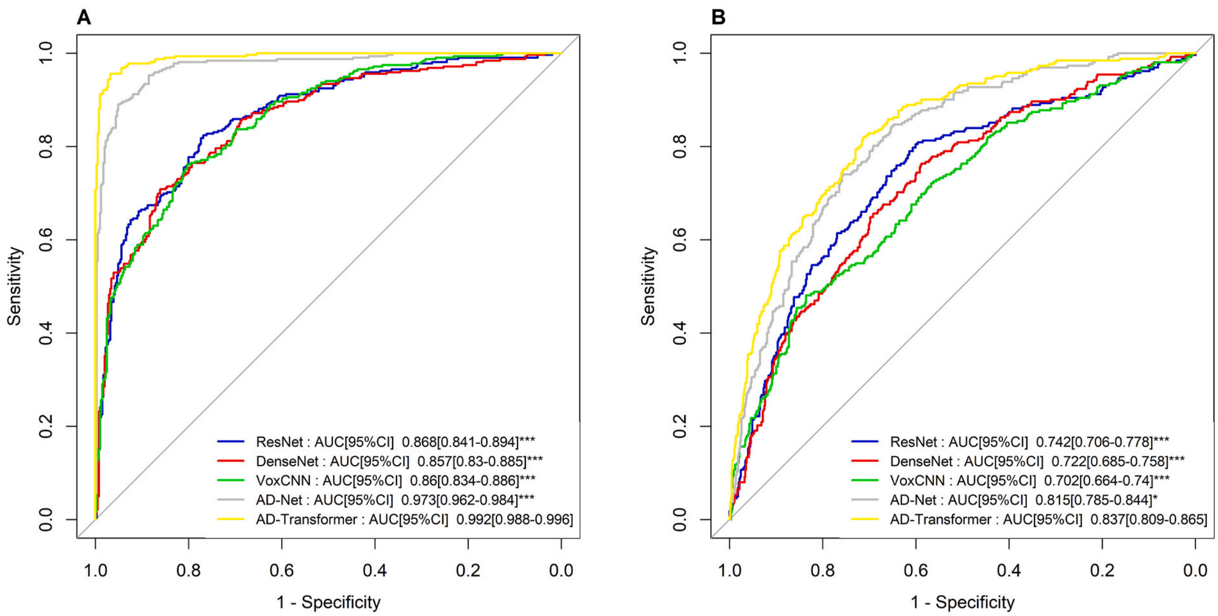

Fig. 2. ROC curve of different models for AD diagnosis (A) and MCI conversion (B).
*P < 0.05,**P < 0.01,***P < 0.001 for difference of AUCs between AD-Transformer and base models.

that integration of the image, clinical and genetic data can effectively improve the performance of AD diagnosis (average AUC: 0.959) and MCI conversion prediction (average AUC: 0.804). This phenomenon confirmed that integrating various multimodal input data can lead to richer representations and improve performance. Finally, the results also show that AD-Transformer achieves the best performance for AD diagnosis and MCI conversion prediction, demonstrating that the token-level attention operations in transformer can aid in capturing and encoding the interconnections among the local patterns of different modalities into the fused representations.

4.2. Model interpretation

Fig. 3 provides gradient-based interpretation results. We see that the image modality played a significant role in the diagnostic process, and its weight was nearly 55.4 % and 55.2 % in AD diagnosis and MCI conversion prediction, respectively. The clinical data was the second most important modality, accounting for roughly 30.7 % and 31.7 % in AD diagnosis and MCI conversion prediction, respectively. In both gradient-based and SHAP based interpretation (Figs. 3 and 4), RAVLT test were the most important clinical item for AD diagnosis and MCI conversion prediction.

To detect the pathological locations, we further calculate the saliency maps to highlight brain regions most related to AD (Fig. 5). For AD assessment, constructing an end-to-end CNN network utilizing the whole brain image may be more appropriate for comprehensive analysis, offering a more holistic perspective of brain information. Patch-based methods can extract information efficiently, and may have the ability to capture local brain features and microstructures (mainly focusing on temporal lobe and frontal lobe), which is important for early lesion detection. Besides, a hybrid CNN-Transformer architecture might emphasize the importance of specific tokens or positions within the input patch sequence, reflecting the attentional focus of the model.

5. Discussion

The study developed a transformer-based multimodal deep learning model based on sMRI, clinical and genetic data. Our AD-Transformer was able to function with flexible combinations of image and non-image data, and outperforms other image-only models and non-unified methods in AD diagnosis and MCI conversion prediction. Compared with the traditional deep learning methods, AD-Transformer was constructed based on a unified framework, which generates diagnostic decisions from multimodal input data and learns holistic multimodal representations.

In our study, in contrast to constructing an end-to-end CNN network using the whole brain image, our proposed patch-based method had better predicting performance. This finding aligns with the results of previous research [33,42,43]. Medical images, especially MRI scans, are often high-resolution and cover a large area. Patch-CNN approaches allow the network to handle large brain images effectively by breaking them down into smaller, more manageable patches, and focusing on local details within the brain MRI images. In contrast, voxel-based methods suffer from the challenge of over-fitting, due to the high dimensionality of features in comparison to the relatively limited number of images available for model training [16]. In our unified framework, the Patch-CNN was responsible for extracting local features from the image data. The extracted features were then processed by transformer layers. The hybrid CNN-Transformer architecture was a powerful combination that leverages the strengths of both CNNs and transformer. This architecture was designed to effectively capture both local and global dependencies within data [44,45].

In AD assessment, previous research has demonstrated that utilizing a multimodal approach often results in enhanced performance compared to relying solely on image data [6,7,46]. Table 4 offered a brief overview of existing models for AD assessment. Tong et al. [47] constructed a graph fusion network that efficiently exploited the similarities from sMRI, PET, CSF and genetic data for AD diagnosis, achieving accuracy and AUC of 0.92 and 0.98, respectively. Liu et al. [48] proposed a multi-task framework including AD classification and clinical score regression, which fused sMRI and clinical data using CNN. The clinical data was directly fed into the fully connected layers as neurons, and reached an accuracy and AUC of 0.94 and 0.99 for AD classification, respectively. Guan et al. [49] have attempted to combine imaging data and clinical data, where the imaging component utilized a CNN, and the clinical information was injected into the model to provide additional context. Moreover, a previous study using deep models for AD classification which integrates image (extracted from CNN), clinical, and genetic data (extracted from auto-encoders) demonstrated that integrating multi-modality data outperforms single modality models

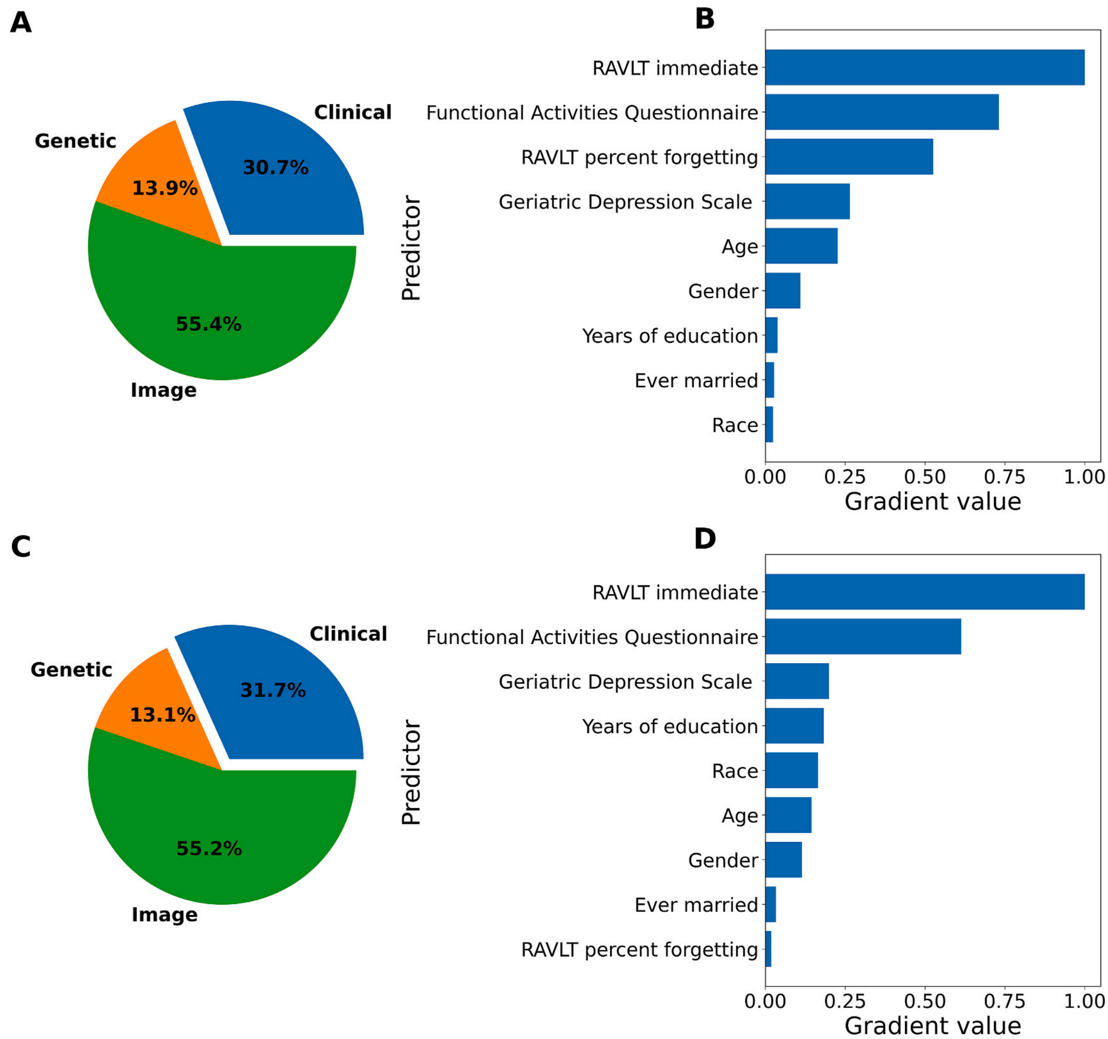

Fig. 3. Gradient-based interpretation on our proposed method.
Gradient value allocated to different types of input, including MRI, genetic and clinical data in (A) AD diagnosis and (C) MCI conversion prediction. Relative importance of clinical items in (B) AD diagnosis and (D) MCI conversion prediction.

[20]. Despite achieving impressive overall performance, these methods overlook a crucial aspect of multi-modal learning: the interaction of modalities. Many existing methods combine multi-modal information in a simplistic manner, merely concatenating features extracted from different modalities. There are more effective approaches to integrating complementary information from modalities, which can enhance the prediction efficiency.

In our study, the unified AD-Transformer framework achieved the best results, with an average AUC of 0.993 and 0.845 in the AD diagnosis and MCI conversion prediction, which was higher than image-only models and the non-unified model (AD-Net) (Table 2). We believed that the performance improvement was closely linked to some capabilities of AD-Transformer. First, AD-Transformer was a unified network, making diagnostic decisions from multimodal input data and learning holistic representations. In comparison, the traditional non-unified approach decomposes the diagnosis problem into several components and may prevent the model from learning holistic and diagnosis-oriented features. Besides, the multi-head mechanism in transformer block enabled the model to attend to diverse relationships during learning, with a focus on capturing different modalities and their interconnections in the input. In contrast, the previous non-unified model directly used the extracted features from different modalities for prediction. By integrating multimodal data, this framework can significantly enhance diagnostic capabilities, ultimately leading to improved patient care and outcomes. In real-world scenarios, the framework could serve as a diagnostic aid for physicians, providing a comprehensive view of a patient's health. This can assist in devising personalized treatment plans, enabling proactive disease management.

In addition to disease prediction, interpretation of the deep learning models is also crucial for multimodal analysis. For AD diagnosis, image data played a significant role in the diagnostic process, followed by clinical and genetic data. In clinical variables, the RAVLT test was identified with the highest weights within the clinical information. The scale evaluates memory function, and is widely recognized for its practicality and ease of use [50]. Similarly, a previous study using deep learning methods based on genetic, imaging, and clinical data also showed that RAVLT test, hippocampus, and amygdala brain areas were distinguished features for AD diagnosis [20]. Our findings align with the conclusions drawn from previous studies on AD [51,52], demonstrating that temporal lobe regions (including the hippocampus) are all highly discriminatory brain regions for the diagnosis of AD. In the early stages of AD, the volume, shape, and texture of the hippocampus are affected, and they have been used as markers of early AD in various studies [53].

5.1. Study limitations and future research directions

Although our AD-Transformer has achieved good performance in AD diagnosis and MCI conversion prediction, it is important to acknowledge

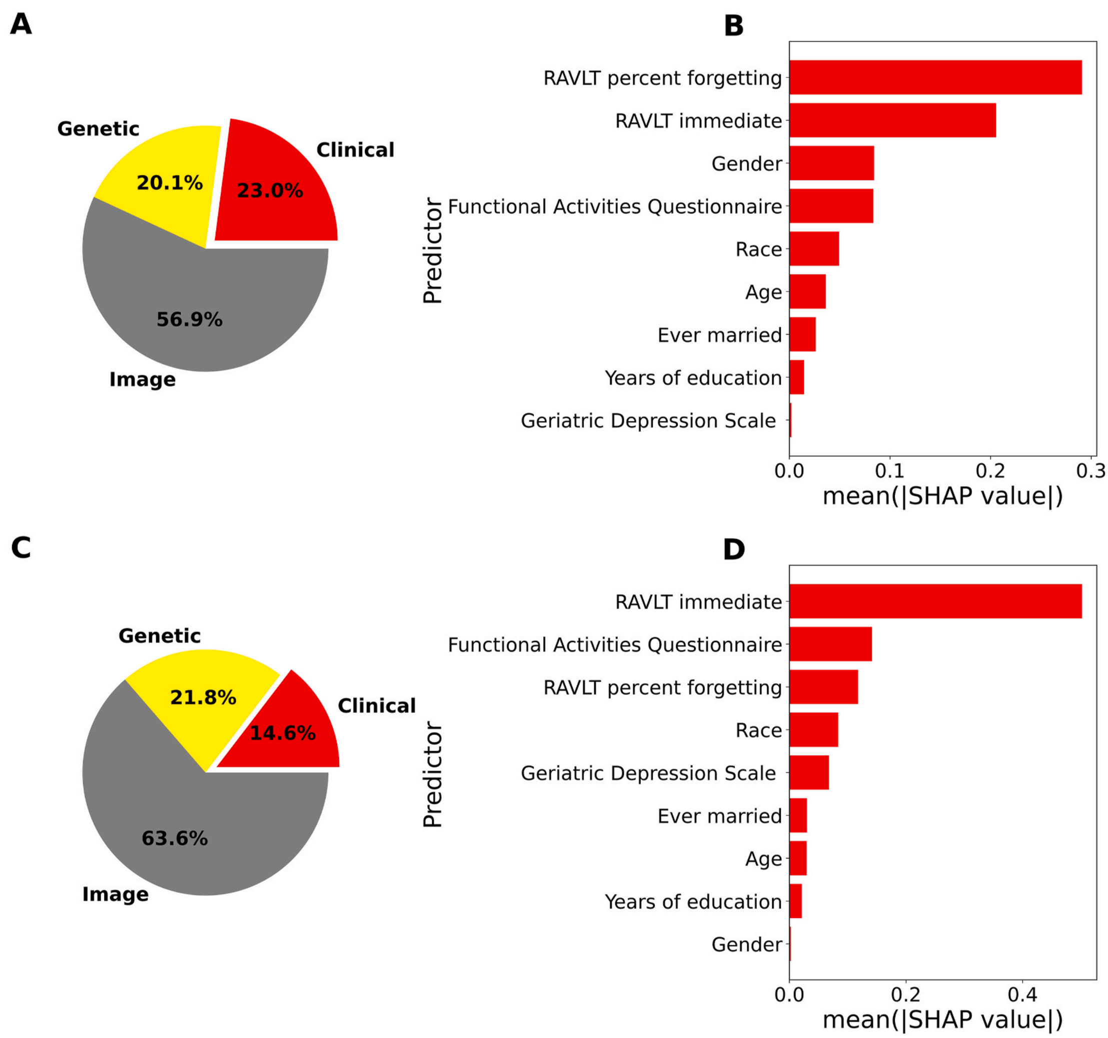

Fig. 4. SHAP-based interpretation on our proposed method
Mean (|SHAP value|) allocated to different types of input, including MRI, genetic and clinical data in (A) AD diagnosis and (C) MCI conversion prediction. Mean (|SHAP value|) of clinical items in (B) AD diagnosis and (D) MCI conversion prediction.

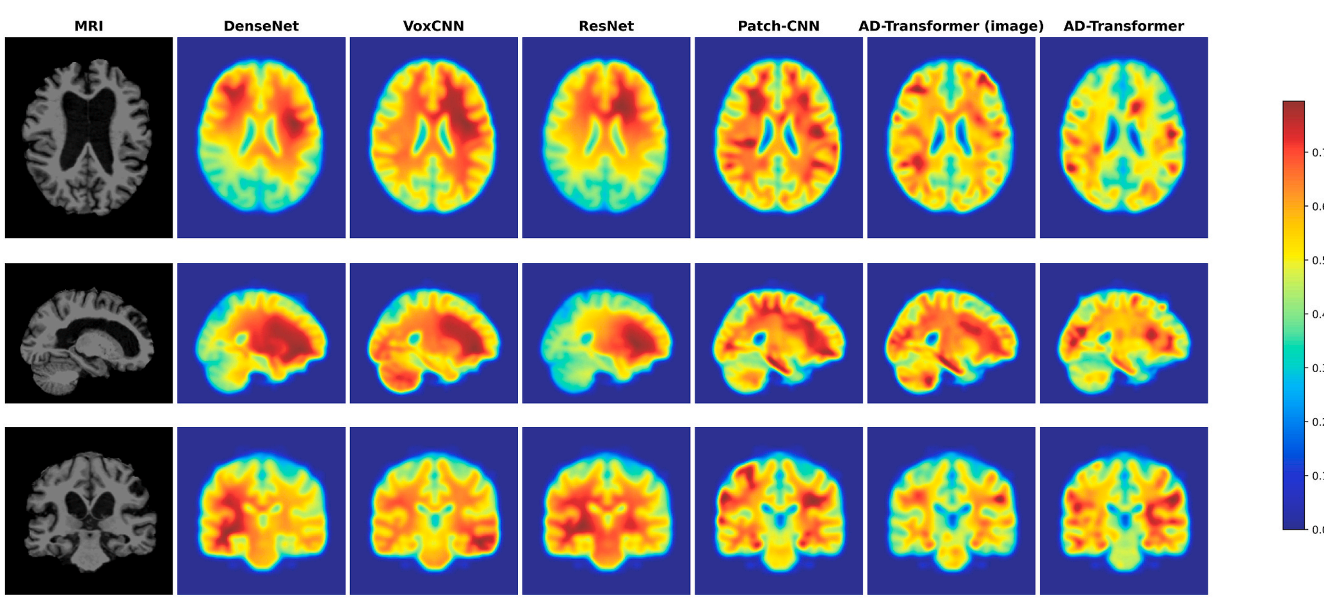

Fig. 5. Visualization results of AD diagnosis based on saliency maps (ID: 005_S_5038).
Image model: DenseNet, VoxCNN, ResNet, Patch-CNN, and AD-Transformer (image). Multimodal model: AD-Transformer.

Table 4
A brief description of the previous studies for AD classification and MCI conversion prediction.

Study	Methods	Modalities	AD vs NC				pMCI vs sMCI			
			ACC	SEN	SPE	AUC	ACC	SEN	SPE	AUC
Korolev et al. [37]	CNN	MRI	0.80	–	–	0.87	0.52	–	–	0.52
Gao et al. [41]	Transformer	MRI	0.91	0.85	0.95	0.94	0.74	0.59	0.82	0.74
Lian et al. [42]	Hierarchical FCN	MRI	0.90	0.82	0.97	0.95	0.81	0.53	0.85	0.78
Pan et al. [54]	Spatially-constrained Fisher representation	MRI + PET	0.94	0.92	0.95	0.97	0.77	0.79	0.77	0.83
Zhang et al. [55]	GNN	MRI + PET + Clinical	0.95	0.93	0.97	0.95	0.77	0.52	0.89	0.71
Tong et al. [47]	Graph fusion network	MRI + PET + CSF + Genetic	0.92	0.89	0.95	0.98	–	–	–	–
Liu et al. [48]	CNN	MRI + Clinical	0.94	0.95	0.93	0.99	–	–	–	–
AD-Transformer	CNN, Transformer	MRI + Clinical + Genetic	0.96	0.96	0.96	0.99	0.75	0.75	0.76	0.85

the following limitations. Our model used genetic, clinical, and imaging data from ADNI database. The paucity of genetic data has prevented the validation of our model on external datasets. Within clinical practice, the lack of racial and ethnic diversity within the ADNI database may impact the generalizability of research findings and models. Moving forward, additional data can be gathered from diverse medical institutions and countries to enhance model's generalizability in real clinical settings.

Besides, this study used the imputation technique to impute missing values in structured data, but failed to consider the problem of deficient image modalities. In the future, we will explore the use of masked modeling techniques to effectively learn from partially observed or incomplete data, thereby enhancing the model's ability to handle missing information.

6. Conclusion

In this work, we proposed a novel transformer-based unified framework for AD diagnosis and MCI conversion prediction based on the sMRI, clinical and genetic data. The transformer block in AD-Transformer is used to effectively capture the interconnections between modalities, and obtain a holistic multimodal representation for disease prediction. The experimental results demonstrate that our AD-Transformer can achieve precise detection capabilities in AD diagnosis and MCI conversion prediction.

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Ethics approval and consent to participate

This research uses data from the Alzheimer's Disease Neuroimaging Initiative (ADNI). All participants provided written informed consent in this survey. All methods of the study were performed in accordance with the prior institutional ethics approval.

Availability of data and materials

This research uses data from the Alzheimer's Disease Neuroimaging Initiative (ADNI), which can be downloaded at https://adni.loni.usc.edu/.

CRediT authorship contribution statement

Qi Yu: Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Qian Ma:** Writing – review & editing, Writing – original draft. **Lijuan Da:** Writing – review & editing, Writing – original draft. **Jiahui Li:** Validation, Investigation. **Mengying Wang:** Validation, Investigation. **Andi Xu:** Validation, Investigation. **Zilin Li:** Validation, Resources, Methodology, Investigation. **Wenyuan Li:** Writing – review & editing, Writing – original draft, Validation, Resources, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that there is no conflict of interest.

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